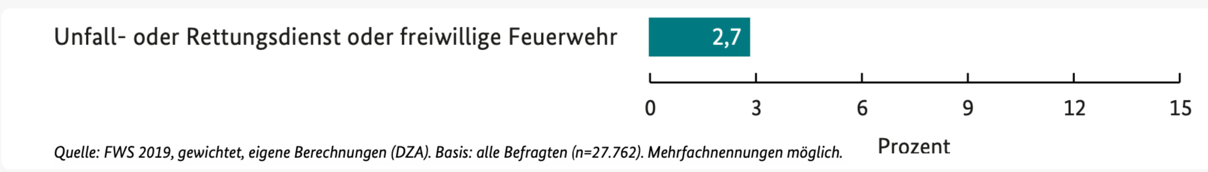
LIFE SCOUT.

Autonomous drone to search for missing persons and monitor their mobility.

Scope

According to Herdel et al. [1], the **emergency sector is the most important area** in which drones are used, accounting for 15.6%. Drones are becoming increasingly popular in emergency situations because they are fast, light, affordable and readily available. As already mentioned, drones measure physiological data in emergency situations. The rescue of people could either be remotely controlled by first responders, which requires a lot of knowledge, or they could autonomously locate people in distress.
DJI, the world's leader in civilian drones and aerial imaging technology reports that in 2017, an estimated 65 people had been rescued with the assistance of drone technology. By June of 2018, that number skyrocketed to 133 people, the world's leader in civilian drones and aerial imaging technology [2] .



The search for missing persons is mainly supported by volunteers. They support rescue situations and are also deployed when official agencies are not yet able to. However, the number of people working as volunteers has declined from 17.1 to 15.7 million. This represents a decrease of 8.2 percent [3]. 2.7% of them are involved in accident and emergency services or volunteer fire departments, which are involved in searching for missing persons.
Drone control requires specialist personnel and its widespread use is very costly, which makes it unattractive for volunteers. How can volunteers and official organizations be efficiently supported with this technology in cases of missing persons?

[1] Herdel, Viviane, Lee J. Yamin, and Jessica R. Cauchard. "*Above and beyond: A scoping review of domains and applications for human-drone interaction*." Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems. 2022.
[2]DroneDeploy. "*How Drones Reshape the Future of Search and Rescue*." DroneDeploy Blog, DroneDeploy, 11.07.2019, https://www.dronedeploy.com/blog/how-drones-reshape-future-of-search-and-rescue.
[3] "*Hilfe in der Krise: Zu wenig Nachwuchs für das Ehrenamt*." BR.de, Bayerischer Rundfunk, 08.10.2022, https://www.br.de/nachrichten/wirtschaft/hilfe-in-der-krise-zu-wenig-nachwuchs-fuer-das-ehrenamt,TJPGQs9 .

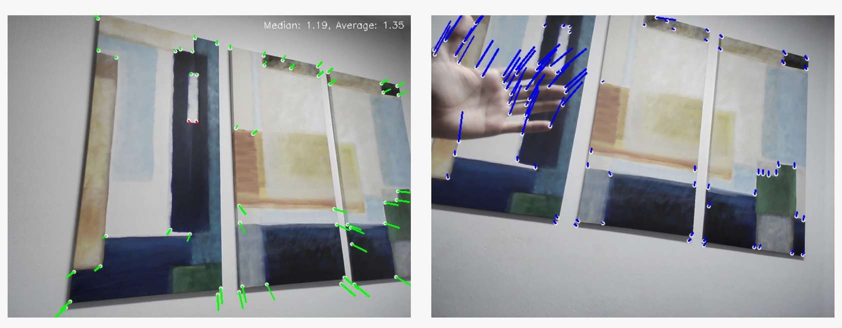
„Oft kommen wir zum Einsatz und müssen dann erst die Gegebenheiten klären, eine Vorabinformation würde sehr viel Zeit sparen. Drohnen wären super, aber große kommen erst bei offiziellen Suchaktionen in Aktion und für den Verein selbst sind sie zu teuer.“

- Jonathan, Volunteer for the rescue service and the water rescue service

Technical Approaches: Evaluated and Discarded

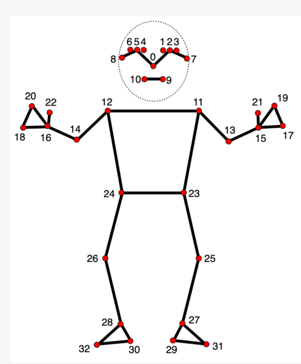
Frame Differencing: Compares consecutive images and "subtracts" them from each other, but is pixel-based and therefore very noisy. Unsuitable with a moving camera as a result of the tests and therefore excluded from the project.

Feature Detection: Finds edges/important points in the image and compares the movement of these points with each other. Calculates the movement of the drone. Better with a moving camera. **Problem:** Depending on the flight direction of the drone, points at the edge of the optical field continue to move



Output of the video stream and the points, that visualizes how different point shift when camera moves ("angle of points" show direction)

Mediapipe Pose model



Coordinate System:

- 2D Coordinates (x/y):
  - 33 landmarks relative to the screen
  - Top-left corner (0,0), bottom-right corner 1,1), center (0.5, 0.5), all values floats
  - Precision value is checked to ensure accuracy
  - These coordinates indicate the landmark positions within the image
- z-Axis:
  - Scaled differently based on the person's distance from the camera
  - Originates from the center of the hips.
  - Indicates proximity to the camera (e.g., which hip point is closer or further)
  - Example: When a person sits, the feet are closer to the camera

Use case

Autonomous drone search for people on suspicion and monitoring of the agility and mobility of the person found

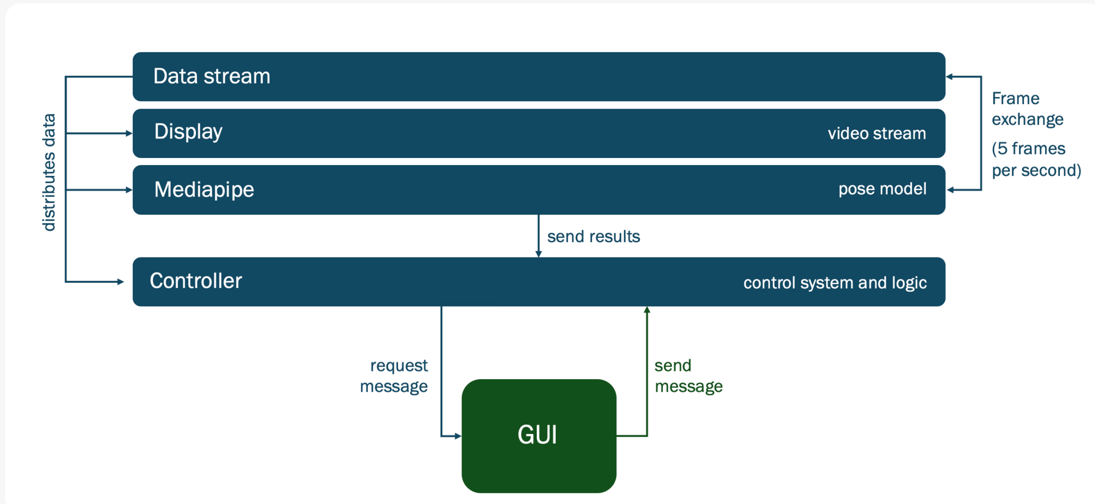
Faster and more efficient localization (on suspicion) of missing persons in areas that are difficult to access or extensive, analysis of the mobility of the person found in order to assess the state of health and determine the need for further rescue measures.

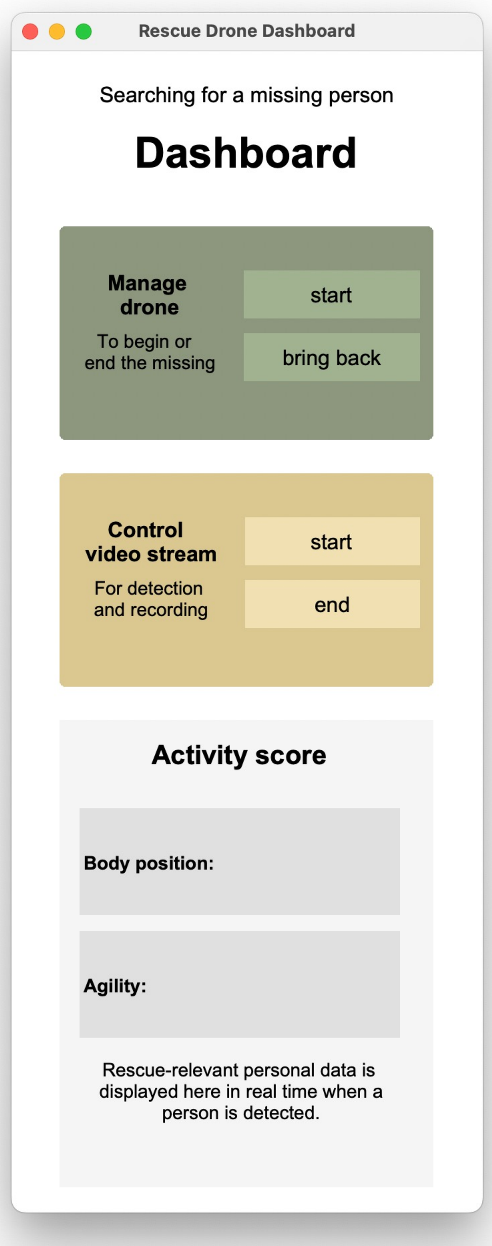
Primarily people from voluntary work, associations etc. (voluntary rescue teams)

- **Quick localization** possible due to time savings (no waiting for official search teams to arrive)
- **Efficiency** through automation (anyone can operate the drone, the mobility of the person is already analyzed and can be evaluated immediately if necessary)
- **Cost savings** (no use of a high-tech drone, program suitable for smaller, low-cost drones)
- **Higher success rate** through early intervention and response (rescue resources or conditions can be calculated in advance)
- **Diverse use** (urban, rural, natural catastrophe)

Backend

4 processes connected through pipelines





Frontend

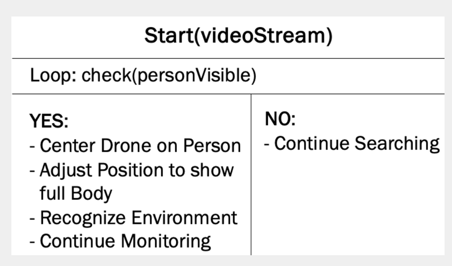
2 windows: dashboard, video stream
Body position: [lying], [standing[, [sitting], arrow for orientation of the person
Agility: [0-1], the higher the more agile

Implementation

Timeline: KW21 – KW26
Due to time and technical problems, the use case could only be completed in parts.

Components droneFlight & videoStream

Either: Video starts and runs smoothly and logic or current workflow of the program is visible in the terminal.

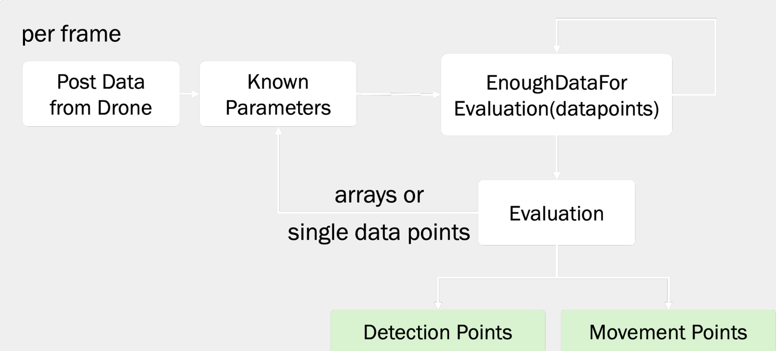


Or: Autonomous flight of the drone in a 1x1m grid

Problem: Process parallelization, frame flow analysis

Approaches to the solution:
- Drone control in parallel
- Request new frames to reduce volume instead of permanent camera recording
- Drone stays stable during detection for pose detection

Component Partial feature detection for mobility monitoring



Detection points: max number evaluation could achieve

Movement points: fitness rating [0-detections points]

End results:

$$\text{Activity score} = \frac{\text{Sum of all movement points}}{\text{Sum of all detection points (of successful evaluations)}}$$

Problem: Drone control and pose evaluation should run quasi-parallel and interact. Control commands shouldn't block or run in a separate thread.

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